

✓ Lab 4 — One-Dimensional & Two-Dimensional Arrays

(Formatted exactly like Lab 2 & 3)

Objectives

By the end of this lab, you will be able to:

- Declare and initialize one-dimensional arrays
 - Access and update array elements
 - Process arrays using loops
 - Find min, max, and average values
 - Declare and process two-dimensional arrays
 - Use nested loops with arrays
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Part 1 — One-Dimensional Arrays (1D)

Question 1 — Array Initialization

Declare an array of size 5 and store the numbers:

10, 20, 30, 40, 50

Then print all elements.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10, 20, 30, 40, 50};

    for (int i = 0; i < 5; i++)
        cout << arr[i] << " ";

    return 0;
}
```

Question 2 — Input + Sum of Array

Ask the user to enter **5 integers**, store them in an array, then print the **sum**.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    int arr[5];
    int sum = 0;

    cout << "Enter 5 numbers: ";
    for (int i = 0; i < 5; i++) {
        cin >> arr[i];
        sum += arr[i];
    }

    cout << "Sum = " << sum;
    return 0;
}
```

Question 3 — Finding the Largest Element

Write a program that reads **6 integers** into an array and prints the **largest** value.

Answer:

```

#include <iostream>
using namespace std;

int main() {
    int arr[6];
    cout << "Enter 6 numbers: ";
    for (int i = 0; i < 6; i++)
        cin >> arr[i];

    int largest = arr[0];
    for (int i = 1; i < 6; i++)
        if (arr[i] > largest)
            largest = arr[i];

    cout << "Largest = " << largest;
    return 0;
}

```

Part 2 — Array Processing Algorithms

Question 4 — Count Positive Numbers

Ask the user for **8 integers** and count how many are **positive**.

Answer:

```

#include <iostream>
using namespace std;

int main() {
    int arr[8];
    int countPos = 0;

    cout << "Enter 8 numbers: ";
    for (int i = 0; i < 8; i++) {
        cin >> arr[i];
        if (arr[i] > 0)
            countPos++;
    }

    cout << "Positive count = " << countPos;
    return 0;
}

```

Question 5 — Average of Array Elements

Input **10 numbers**, find the **average**, and print it.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    int arr[10];
    int sum = 0;

    cout << "Enter 10 numbers: ";
    for (int i = 0; i < 10; i++) {
        cin >> arr[i];
        sum += arr[i];
    }

    cout << "Average = " << sum / 10.0;
    return 0;
}
```

Part 3 — Two-Dimensional Arrays (2D)

Question 6 — Input and Display a 2×3 Matrix

Ask the user to enter values for a **2-row, 3-column** matrix, then print the matrix.

Answer:

```

#include <iostream>
using namespace std;

int main() {
    int matrix[2][3];

    cout << "Enter 6 numbers for a 2x3 matrix:\n";
    for (int r = 0; r < 2; r++)
        for (int c = 0; c < 3; c++)
            cin >> matrix[r][c];

    cout << "Matrix:\n";
    for (int r = 0; r < 2; r++) {
        for (int c = 0; c < 3; c++)
            cout << matrix[r][c] << " ";
        cout << endl;
    }

    return 0;
}

```

Question 7 — Sum of Each Row

Given a **3×3 matrix**, compute and print the sum of each row.

Answer:

```

#include <iostream>
using namespace std;

int main() {
    int matrix[3][3];

    cout << "Enter 9 numbers for a 3x3 matrix:\n";
    for (int r = 0; r < 3; r++)
        for (int c = 0; c < 3; c++)
            cin >> matrix[r][c];

    for (int r = 0; r < 3; r++) {
        int rowSum = 0;
        for (int c = 0; c < 3; c++)
            rowSum += matrix[r][c];

        cout << "Sum of row " << r << " = " << rowSum << endl;
    }

    return 0;
}

```

Part 4 — Searching Arrays

Question 8 — Search for a Value

Ask the user for **10 numbers**, then ask for a value **x** and check if **x** exists in the array.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    int arr[10];
    cout << "Enter 10 numbers: ";
    for (int i = 0; i < 10; i++)
        cin >> arr[i];

    int x;
    cout << "Enter value to search: ";
    cin >> x;

    bool found = false;
    for (int i = 0; i < 10; i++)
        if (arr[i] == x)
            found = true;

    if (found)
        cout << "Value found.\n";
    else
        cout << "Value not found.\n";

    return 0;
}
```